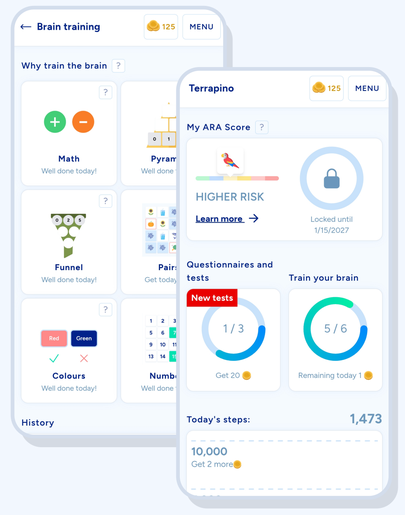
Terrapino

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Digital Cognitive Training and Dementia Risk Monitoring: Real-World Evidence from the Terrapino Mobile App



Jan Pavlik[4], Katerina Sheardova[3a,3b], Martin Valis[5], Jiri Cerman[2], Ross Andel[1], Jakub Hort[2]

[1]Edson College of Nursing and Health Innovation, Arizona State University, Phoenix, AZ, United States. [2]Department of Neurology, Second Faculty of Medicine, Charles University, Motol and Homolka University Hospital, Prague, Czechia. [3a]International Clinical Research Center, St. Anne's University Hospital, Brno, Czechia. [3b]Neuro Health Centrum, Brno, Czechia. [4]AlzheimerChain Foundation, Prague, Czechia. [5]Research Institute for Biomedical Science, Hradec Kralove, Czechia.

A gamified brain-health app reaches the people dementia prevention needs to reach: two in three of 38,411 real-world users are aged 50–80.

38,411 users, 3 years of real-world app data (Czech & English)

67% aged 50–80, the core target group for prevention

36% of 18,733 risk-assessed users in an increased-or-higher risk category

41% below-normal memory score on the Harrison BASIC test

1. Background

- Prevention is a major factor in reducing disease onset; modifying 14 risk factors could prevent or delay up to 45% of dementia cases worldwide.[1]
- In addition to activation through specific cognitive exercises, physical activity (such as walking) is crucially important, as maintaining physical activity protects against dementia.
- Prevention needs tools that reach people **before symptoms**, at population scale: risk screening, cognitive testing and lifestyle support.
- Mobile apps can do this, but **keeping users engaged** long term is the main barrier in digital health.

2. Objective

- To evaluate user uptake and long-term engagement with the Terrapino mobile app.
- To characterize the data streams generated by the platform.
- To demonstrate how the app supports dementia risk screening, cognitive training, and healthy lifestyle habits.

3. Methods

- **Design:** retrospective analysis of anonymized real-world data from **38,411** Czech- and English-speaking users over 3 years.
- **What the app measures and trains:**
  - **ARA score:** Alzheimer's Risk Assessment questionnaire on modifiable risk factors, physical activity and diet.
  - **Harrison BASIC:** repeatable episodic-memory test (Brief And Simple Index of Cognition, with Prof. John Harrison).
  - **Emotion recognition** task (social cognition).
  - **Daily training:** 7 cognitive mini-games and a step tracker for walking.
- **Motivation:** points for activity, redeemable for partner offers.
- **Outcomes:** uptake, engagement, risk categories, cognitive test results.

4. Results

Fig 1. Two in three users are aged 50–80

The app reaches the age group at increased risk of cognitive decline, not only younger, tech-savvy users.

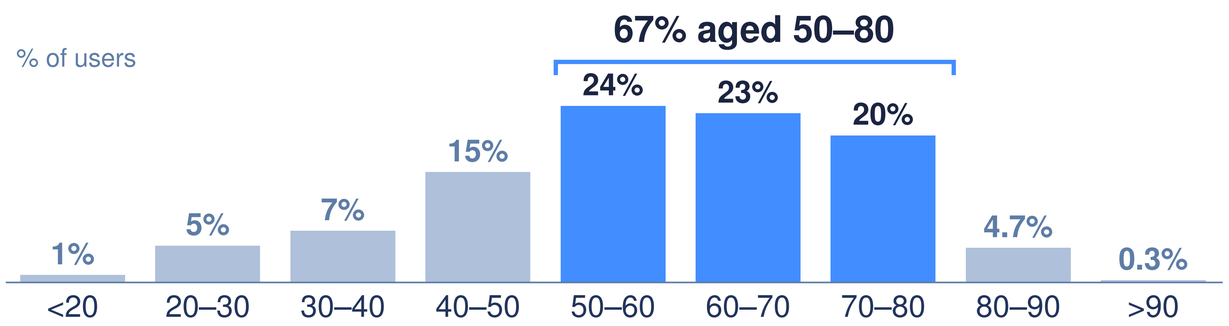
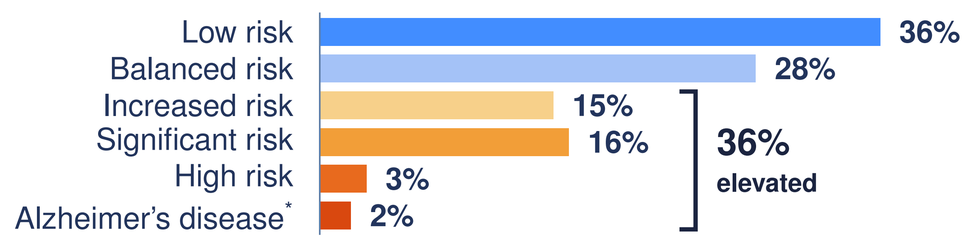


Fig 2. One in three risk-assessed users has elevated risk

48.8% of users (n = 18,733) completed the ARA questionnaire at least once.



*Name of the highest ARA questionnaire category; not a clinical diagnosis.

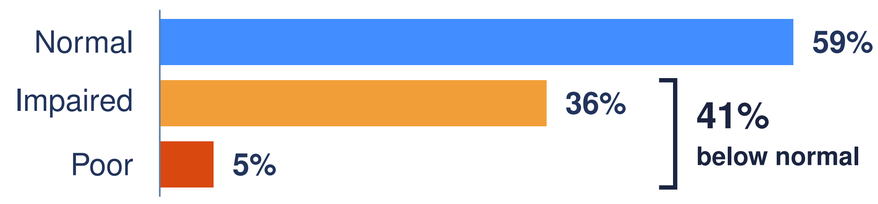
Fig 3. Users and engagement

10:33 min mean session length, highly active users

1,038 daily active users of the 7 games (last 6 months)

Fig 4. 41% scored below normal on a memory test

Harrison BASIC, ≈4,500 users tested. A screening result in a self-selected sample, to be confirmed clinically.



5. Conclusion

- **Target population reach:** a gamified app reached 38,411 users, two thirds of them in the 50–80 age group that prevention targets.
- **Risk is visible early:** one in three risk-assessed users had an increased-or-higher risk profile, and 41% of tested users scored below normal on memory.
- **Users engage:** highly active users spend over 10 minutes per session; points-based rewards are a practical engagement lever.

Next steps

- Longitudinal analysis of repeated Harrison BASIC scores and long-term retention.
- Predictive models and personalized lifestyle recommendations.
- Physician reports that bring app data into clinical care.
- More language versions.

App download

Access poster PDF

Medical device: Terrapino (software), Risk class I, according to the Annex VIII, Rule 11 Regulation (EU) 2017/745 of the European Parliament and of the Council on medical devices.

Reference: 1. Livingston G, Huntley J, Liu K et al. Dementia prevention, intervention, and care: 2024 report of the Lancet standing Commission. The Lancet, 2024; 404, 572–628.